Quandles and second relative homotopy groups

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Quandles

A ***quandle*** is a set Q equipped with two binary operations $\triangleright$ and $\triangleright^{-1}$ satisfying the following axioms:

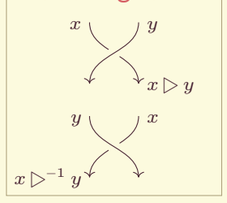
(i) Idempotency: $x \triangleright x = x$.
(ii) Inverses: $(x \triangleright y) \triangleright^{-1} y = x = (x \triangleright^{-1} y) \triangleright y$.
(iii) Self-distributivity: $(x \triangleright y) \triangleright z = (x \triangleright z) \triangleright (y \triangleright z)$.

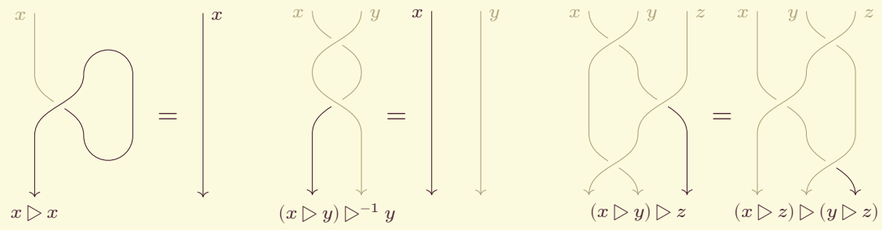
The two operations determine each other uniquely.

Reidemeister moves

The quandle axioms can be interpreted as invariance under the Reidemeister moves of knot diagrams:

Labelling rules:





Group conjugation

Given a group G, define its ***conjugacy quandle*** $\operatorname{Conj} G$ as the same underlying set endowed with the operations

$$g \triangleright h := g^h = h^{-1}gh, \qquad g \triangleright^{-1} h := {}^h g = hgh^{-1}.$$

That is, we forget the group operation and only remember the action by conjugation.

Mnemotechnic: *"The base [of the triangle] points towards the base [of the conjugation]".*

Conversely, given a quandle Q, define its ***associated group*** $\operatorname{As} Q$ by the presentation

$$\langle \{e_q \mid q \in Q\} \mid \{e_{q \triangleright p} = e_p^{-1} e_q e_p \mid q, p \in Q\} \rangle.$$

These are adjoint: giving a group map $\operatorname{As} Q \to G$ is the same as giving a quandle map $Q \to \operatorname{Conj} G$.

The fundamental quandle

Let $L \hookrightarrow M$ be a smooth codimension 2 embedding of oriented manifolds and $N(L)$ a tubular neighbourhood.

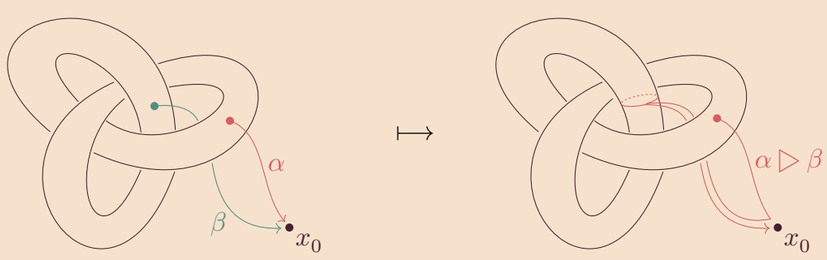
The ***fundamental quandle*** of the embedding at a point $x_0 \in M \setminus N(L)$ is the set

$$\Gamma(L \hookrightarrow M; x_0) := \{\text{maps } ([0,1], \{0\}; 1) \to (M, \partial N(L); x_0) \text{ up to relative homotopy}\}.$$

That is, the endpoint of each path is fixed, but the starting point can wander freely through $\partial N(L)$.
The quandle operation is given by

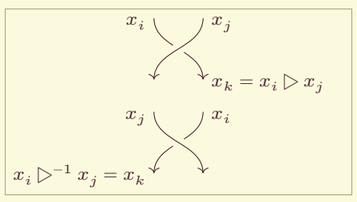
$$\alpha \triangleright \beta := \alpha * \beta^{-1} * m_\beta * \beta,$$

where m_β is the loop around $\partial N(L)$ at the point β_0. The direction of m_β is determined by the orientation.



The knot quandle

For links $L \hookrightarrow \mathbb{S}^3$, the fundamental quandle admits the ***Wirtinger presentation***.



Take a link diagram and label each arc with letters $x_1, \ldots, x_n$.
At each crossing, consider the relation $x_k = x_i \triangleright^{\pm 1} x_j$.

$$\Gamma(L \hookrightarrow \mathbb{S}^3) \cong \langle x_1, \ldots, x_n \mid \{x_k = x_i \triangleright^{\pm 1} x_j\} \rangle.$$

So the arcs become generators and the crossings become relations.

The original Wirtinger presentation concerns the fundamental group of the link complement, where we have

$$\pi_1(\mathbb{S}^3 \setminus L) \cong \langle x_1, \ldots, x_n \mid \{x_k = x_j^{\mp 1} \cdot x_i \cdot x_j^{\pm 1}\} \rangle.$$

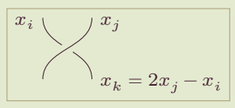
In particular, $\operatorname{As} \Gamma(L \hookrightarrow \mathbb{S}^3) \cong \pi_1(\mathbb{S}^3 \setminus L)$. But the fundamental quandle is a much stronger invariant.

Theorem [1, § 16; cf. 2, § 5]. The fundamental quandle $\Gamma(L \hookrightarrow \mathbb{S}^3)$ of an oriented knot L in the 3-sphere contains exactly the same information as the triple (G, P, m) formed by knot group $G := \pi_1(\mathbb{S}^3 \setminus L)$, the peripheral subgroup $P := \pi_1(\partial N(L)) \cong \mathbb{Z}^2 \hookrightarrow G$, and a choice of meridian $m \in P$.

That is to say, the fundamental quandle distinguishes all oriented knots up to simultaneous application of mirror image and orientation reversal. In particular, it is a complete invariant of unoriented knots.

Knot colorability as a quandle invariant

In general it is quite difficult to prove whether any two presented quandles are isomorphic.
Therefore, we often want to look for simpler invariants encoded in the knot quandle.
As an example, let's look at the classical invariant of Fox n-colorability. Fix an odd number n.



An ***n-coloring*** of an unoriented knot is a labelling of its diagram by elements of $\mathbb{Z}_{/n}$ such that at each crossing one has $2x_j = x_i + x_k$. Equivalently, $x_k = 2x_j - x_i$.

A knot is called ***n-colorable*** if it admits a non-constant n-coloring. The unknot is not n-colorable for any n.

Fixing a base arc and setting its label to zero (by substracting everywhere), we can identify n-colorings (up to a constant) with representations $\rho \colon \pi_1(\mathbb{S}^3 \setminus L) \to D_{2n}$ of the knot group in the dihedral group of order $2n$.

Define the ***dihedral quandle*** $D_n^{\mathbb{Q}}$ of order n as the subquandle of $\operatorname{Conj} D_{2n}$ consisting of the reflections:

$$D_n^{\mathbb{Q}} = \{0, \ldots, n-1\} = \mathbb{Z}_{/n}, \qquad x \triangleright y = 2y - x = x \triangleright^{-1} y.$$

Now define an ***n-coloring*** of an oriented knot as a morphism $\rho \colon \Gamma(L \hookrightarrow \mathbb{S}^3) \to D_n^{\mathbb{Q}}$.
This recovers our prior definition and generalizes to colorings by an arbitrary quandle Q. This is the starting point for more refined knot invariants such as certain state-sum invariants arising from quandle cohomology.

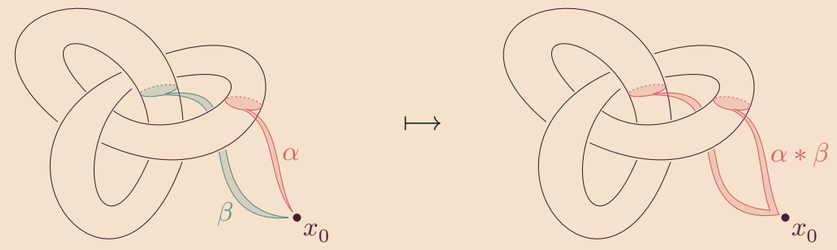
Relative homotopy groups

Given a pair of spaces $(X, X \setminus A)$, its ***relative homotopy groups*** at a point $x_0 \in X \setminus A$ are the sets

$$\pi_n(X, X \setminus A; x_0) := \{\text{maps } (D^n, \partial D^n; *) \to (X, X \setminus A; x_0) \text{ up to relative homotopy}\}.$$

For $n \geq 2$ we can endow these with a natural group structure, which is abelian for $n \geq 3$.

Example: $\pi_2(\mathbb{S}^3, \mathbb{S}^3 \setminus L)$ for a knot $L \hookrightarrow \mathbb{S}^3$. This coincides with $\pi_1(\mathbb{S}^3 \setminus L)$, as we will see.



Alternative definition via the homotopy fiber of the inclusion

Consider the homotopy fiber of the pointed inclusion $i \colon (X \setminus A; x_0) \hookrightarrow (X; x_0)$, which we can construct as the space of paths in X from an unfixed point of $X \setminus A$ to the fixed basepoint $x_0 \in X \setminus A$:

$$\operatorname{Hofiber}(i) := \{(a, \gamma) \in (X \setminus A) \times X^I \mid \gamma_0 = i(a) \text{ and } \gamma_1 = x_0\} \cong \{\gamma \in X^I \mid \gamma_0 \in X \setminus A \text{ and } \gamma_1 = x_0\}.$$

Theorem. The homotopy groups of this space are precisely the relative homotopy groups of the pair:

$$\pi_n(\operatorname{Hofiber}(i)) \cong \pi_{n+1}(X, X \setminus A).$$

Furthermore, we have a fibration $\operatorname{Hofiber}(i) \hookrightarrow \mathrm{E}_i \twoheadrightarrow X$, where the middle space is the mapping path space of the inclusion; namely, it is the space of paths in X that start at an unfixed point in $X \setminus A$:

$$\mathrm{E}_i := \{(a, \gamma) \in (X \setminus A) \times X^I \mid \gamma_0 = i(a)\} \cong \{\gamma \in X^I \mid \gamma_0 \in X \setminus A\}.$$

This space deformation retracts to $X \setminus A$, where the inclusion $X \setminus A \hookrightarrow \mathrm{E}_i$ takes a point to its constant path.

The long exact sequence of relative homotopy groups

Given a pair of spaces $(X, X \setminus A)$ and a basepoint $x_0 \in X \setminus A$, we can construct a long exact sequence

$$\cdots \xrightarrow{\partial} \pi_2(X \setminus A) \xrightarrow{i_*} \pi_2(X) \xrightarrow{j_*} \pi_2(X, X \setminus A) \xrightarrow{\partial} \pi_1(X \setminus A) \xrightarrow{i_*} \pi_1(X) \xrightarrow{j_*} \pi_1(X, X \setminus A) \xrightarrow{\partial} \pi_0(X \setminus A) \xrightarrow{i_*} \pi_0(X)$$

where the ∂ map restricts a disc $(D^n, \partial D^n) \to (X, X \setminus A)$ to its boundary $\partial D^n \to X \setminus A$.
The other maps are induced from the inclusions $i \colon X \setminus A \hookrightarrow X$ and $j \colon (X, \{x_0\}; x_0) \hookrightarrow (X, X \setminus A; x_0)$.

The exactness of the sequence can be proven directly, but it can also be constructed as the homotopy long exact sequence of the fibration $\operatorname{Hofiber}(\iota) \hookrightarrow \mathrm{E}_\iota \twoheadrightarrow X$ that we mentioned earlier.

The group associated to the fundamental quandle

Theorem [2, Prop. 3.2] The associated group of the fundamental quandle of a smooth codimension 2 embedding $L \hookrightarrow M$ is canonically isomorphic to the second relative homotopy group of the pair $(M, M \setminus L)$:

$$\Gamma(L \hookrightarrow M) \cong \pi_2(M, M \setminus L).$$

In particular, if the ambient space M is 2-connected (i.e. $\pi_1(M) = \pi_2(M) = 0$), by the prior long exact sequence we have $\pi_2(M, M \setminus L) \cong \pi_1(M \setminus L)$ and therefore

$$\operatorname{As} \Gamma(L \hookrightarrow M) \cong \pi_1(M \setminus L).$$

Remember that this was already the case for links in the 3-sphere (by the Wirtinger presentations).

Crossed modules

The map $\partial \colon \pi_2(X, X \setminus A) \to \pi_1(X \setminus A)$ has two nice properties that make it part of a ***crossed module***.

First, note that we have a canonical action $\pi_1(X \setminus A) \curvearrowright \pi_2(X, X \setminus A)$ that we will denote $(\gamma, \alpha) \mapsto \alpha^\gamma$.
We can picture this action as "pinching and dragging" a given disc α along the loop γ.

(i) The map ∂ is $\pi_1(X \setminus A)$-*equivariant*: $\partial(\alpha^\gamma) = \partial(\alpha)^\gamma := \gamma^{-1} * \partial(\alpha) * \gamma$.
(ii) The *Peiffer identity* is satisfied: $\alpha^{\partial(\beta)} = \partial(\beta^{-1} * \alpha * \beta)$.

A (right) ***crossed module*** is a quadruple (G, H, ∂, η) of two groups G and H, a group morphism $\partial \colon H \to G$, and a (right) group action $\eta \colon G \curvearrowright H$ satisfying the two prior axioms.

Crossed modules are equivalent to ***strict 2-groups***: strict 2-groupoids on a single object.
The 1-morphisms correspond to elements of G, and the 2-morphisms correspond to elements of $G \times H$.
The elements of H are 2-morphisms $h \colon e \to \partial(h)$ that start at the identity, with the boundary map ∂ acting as the target map. The different ways of composing are encoded in the group structures and in the action η.

Augmented quandles

The quandle analogue to a crossed module is an ***augmented quandle***: a quadruple (Q, G, ∂, η) where Q is a set, G is a group, $\partial \colon Q \to G$ is a function, and $\eta \colon G \curvearrowright Q$ is a group action satisfying the following two axioms.

(i) The map ∂ is G-equivariant: $\partial(x^g) = \partial(x)^g := g^{-1} \partial(x) g$.
(ii) We have "idempotency": $x^{\partial(x)} = x$.

This induces a quandle structure on Q, by setting $x \triangleright y := x^{\partial(y)}$.

For a codimension 2 embedding $L \hookrightarrow M$, the ***fundamental augmented quandle*** is defined by

$$Q := \Gamma(L \hookrightarrow M), \qquad G := \pi_1(M \setminus L), \qquad \partial(\alpha) := \alpha^{-1} * m_\alpha * \alpha, \qquad \alpha^\gamma := \alpha * \gamma.$$

Fundamental augmented quandles (and fundamental crossed modules) satisfy a Seifert–van Kampen theorem, while plain quandles (and relative homotopy groups) do not; see [1, § 13].
However, the pushouts involved are pushouts *of augmented quandles*, which can be hard to compute.

[1] D. Joyce. "A Classifying Invariant of Knots, the Knot Quandle". 1982. DOI: 10.1016/0022-4049(82)90077-9.

[2] R. Fenn and C. Rourke. "Racks and Links in Codimension Two". 1992. DOI: 10.1142/S0218216592000203.